YP-RAKSHA

Real-time AI Knowledge Screening & Hazard Audit

System Manual

Field	Value
Product name used in this document	YP-RAKSHA
Name in the running software	Gateway (default). Brand mark: Raksha.
Subtitle in the product	Real-time AI Knowledge Screening & Hazard Audit
Organisation default	YesPanchi Group of Companies
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Classification	Internal
Audience	System administrators, DevOps, IT operations, and security teams
Source of truth	This Git repository. Not a certification claim.

The repository default product string is Gateway, set by GATEWAY_PRODUCT_NAME. The brand string defaults to Raksha. This document set uses the requested product name YP-RAKSHA for the same implementation.

This document describes behaviour found in source code, configuration samples, and existing Markdown guides. Where a feature is partial or absent, the text says so. It does not claim ISO certification, complete hazard detection, or guaranteed AI safety.

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1. System overview

Operators run one gateway service, one PostgreSQL database, and optionally a network proxy. Employees who need browser controls install Gateway Guard. Administrators use the workspace UI. This manual separates steps that match the repository from practices that are recommended but not encoded as a single script.

2. Prerequisites

Item	Evidence in repository
Gateway	Go service. Docker Compose is the packaged path.
Database	DB_TYPE default postgres. Port comes from DB_PORT.
UI dev	Node tooling in ui/. Production UI is embedded in the gateway binary.
Windows Guard build	Windows host and Inno Setup for Setup.exe.
macOS Guard native rebuild	Must be built on a Mac. A Windows host cannot produce a new signed Mach-O.
Directory	Outbound LDAP or LDAPS to a domain controller, or inbound HTTPS from the IdP to /scim/v2.

Hardware sizing is not fixed in code. Existing docs discuss production sizing. Treat those as planning guidance and verify on your workload. This manual does not invent CPU or RAM minimums beyond what you measure.

3. Installation with Docker Compose

Verified shape of deployment, from docker-compose.yml:

1. Clone the repository onto the server.
2. Create a .env file. Set CONTAINER_NAME, APP_PORT, SERVER_DOMAIN, DB_HOST, DB_PORT, DB_NAME, DB_USER, DB_PASSWORD, ADMIN_EMAIL, ADMIN_PASSWORD, and GATEWAY_GUARD_SECRET. Use your own values. Do not copy secrets into tickets.
3. Optional brand variables: GATEWAY_PRODUCT_NAME, GATEWAY_PRODUCT_SUBTITLE, GATEWAY_COMPANY_NAME. Defaults are Gateway, the hazard-audit subtitle, and YesPanchi Group of Companies.
4. From the repository root run: docker compose up -d --build
5. Expected result: container gateway_tech (or your CONTAINER_NAME) is running and listens on APP_PORT.
6. Open https://<SERVER_DOMAIN> through your reverse proxy, or http://<host>:<APP_PORT> on a trusted network.
7. Sign in with ADMIN_EMAIL and the admin password. Change it after first login if policy requires.

Optional network proxy, only if you intend office PAC inspection on the server:

```
docker compose --profile network-proxy up -d
```

The compose comment states the production employee path is the Guard program on each laptop, not the shared proxy.

4. Configuration reference

Only variables referenced by docker-compose.yml or the Guard and gateway code should be treated as real. Common ones:

Variable	Purpose
APP_PORT	Gateway listen port.
APP_HOST	Bind address. Compose default 0.0.0.0.
SERVER_DOMAIN	Public origin used in links and Guard config.
DB_*	PostgreSQL connection.
DB_SSL_MODE	Default disable in compose. Set require when the database demands TLS.
ADMIN_EMAIL / ADMIN_PASSWORD	Bootstrap administrator.
GATEWAY_GUARD_SECRET	Shared secret for Guard downloads and agent calls. Required.
GATEWAY_GUARD_REQUIRE_SECRET	Default 1. Keep it on.
LOG_LEVEL / LOG_STYLE	Process logs. Style default json.

Local Guard ports, from operational notes in the repository, are 18103 for the MITM proxy and 18195 for the PAC server, both on 127.0.0.1. Do not publish those ports to the office network.

5. Provider and model configuration

8. Sign in as admin.
9. Open Models, then Model Providers.
10. Add a provider that exists in core/providers and paste the vendor key into the secret field. The key is stored by the gateway.
11. Confirm the model appears in Model Catalog.
12. Create a virtual key under Governance, Virtual Keys. Attach budgets if you need spend limits.
13. Call the gateway with Authorization: Bearer <VIRTUAL_KEY>. Expected result: a chat completion returns when the provider accepts the model name.

If the provider returns an authentication error, the gateway is up and the vendor rejected the key. Rotate the key in the provider console and update the stored key. Do not send the master key to laptops.

6. Database

14. Create a PostgreSQL database and user matching DB_NAME and DB_USER.
15. Grant that user rights on the database.
16. Point DB_HOST and DB_PORT at the instance. Use 5432 unless you front the database with a pooler.
17. Start the gateway. GORM migrations in framework/configstore run as part of startup.
18. Verify by signing in. A successful admin login means the user tables were created and the bootstrap admin was accepted.

Backup and restore commands are operational. Take a logical dump with pg_dump using your own connection string. Restore with psql or pg_restore. The repository Markdown guides describe schedules. Confirm the schedule on your host. This manual does not claim an already installed cron job.

7. Gateway Guard installation

7.1 Windows

19. In Browser AI, Setup, download Gateway_Guard_Windows.zip.
20. Extract the zip fully. Do not run Setup from inside the zip.
21. Run Gateway_Guard_Setup.exe. Leave start at login enabled if that is your standard.
22. Expected result: Gateway Guard appears in Apps, and Browser AI, Guard Agents shows the laptop after it heartbeats.

Update: run Update_Gateway_Guard.ps1 beside a config file that contains backend_url and guard_secret. The script downloads /api/browser-ai/setup/download-windows.zip, extracts Gateway_Guard_Setup.exe, and installs it silently. Current release version advertised with the 1.1.20 packages is 1.1.20.

7.2 macOS

23. Download Gateway_Guard_macOS.zip from Browser AI, Setup.
24. Or run Update_Gateway_Guard_macOS.command. It reads gateway_guard_config.json, downloads the macOS zip, and copies Gateway_Guard.app into the user or system Applications folder.
25. Expected result: the app opens and the agent heartbeats. Version files inside the current zip are 1.1.20.

Unsupported as a current 1.1.20 installer: Gateway_Guard_Setup.pkg in the release folder was older at review time and is not inside the macOS zip. Do not distribute that pkg as the 1.1.20 update. Build a new pkg on a Mac if you need a package installer.

8. Users, roles, and policies

26. Open Governance, Users.
27. Create a user with email. The server generates a temporary password and emails it. First login must set a new password.

28. Role User: choose team, virtual key, workspace pages, and prompt repositories. Unchecked pages stay hidden, except Prompt Repository remains available for the user role.
29. Role Sub-admin: choose workspace pages. Those pages show in the sidebar. Write access still follows the sub_admin role in Roles and Permissions, not every tick on the user form.
30. Role Admin: full workspace and API access.
31. Open Governance, Roles and Permissions to edit the All Users or All Sub Admins policy if the whole role must change.

8.1 Browser screening rules

32. Open Browser AI, Target Websites. Add hosts you intend to inspect.
33. Open Guard Rules. Add a regex rule or an AI Guard Bot rule. Set the action to block, redact, or warn.
34. Confirm a test prompt from a managed browser appears in Prompt Logs.
35. If it does not, the site is not a target, the PAC is not applied, or the agent is offline.

There is no global switch in code that proves every AI site on the internet is screened. Only configured targets are in scope.

9. Directory connection

9.1 Cloud SCIM

36. Open Governance, User Provisioning. Choose Cloud Identity Providers.
37. Select Okta, Entra, or Keycloak. Enable SCIM. Save.
38. Copy the base URL ending in /scim/v2 and the bearer token into the IdP. Store the token in a password manager.
39. Assign a test user. Expected result: the person appears under Governance, Users with role User.
40. Push a group if you use groups. Expected result: a team with that display name.

9.2 On-premises Active Directory

41. Open the Microsoft Active Directory tab.
42. Set the server URL (ldap:// or ldaps://), domain, base DN, and bind account. Prefer LDAPS. Avoid insecure certificate skip except in a lab.
43. Save, then Test connection. Expected result: a success message and a sample of users.
44. Sync Users Now. New accounts are role User, approved, and emailed a temporary password.
45. If a username already exists, that account is left unchanged.
46. Set a default team in the SCIM configuration if new AD users must land in a team. Otherwise they have no team.
47. After sync, edit each user who needs more than Prompt Repository: virtual key and workspace pages.

10. Logging, health, backup, and upgrades

- Gateway logs follow LOG_LEVEL and LOG_STYLE.
- Product audit logs are the Audit Logs workspace page, when the caller is allowed to open it.

- LLM Logs and Browser AI Prompt Logs are the operational record of AI use.
- Health: process up, admin login works, a virtual-key chat returns, a Guard agent heartbeat appears.
- Upgrade the gateway by rebuilding the image and recreating the container. Take a database dump first.
- Upgrade Guard with the Windows or macOS update script, or by shipping the matching zip from Setup.

11. Troubleshooting

Symptom	Check
Admin login fails	ADMIN_EMAIL, password, and database connectivity. Read gateway logs.
Provider 401	Stored vendor key and model name. The gateway itself is reachable.
User sees only Prompt Repository	allowed_sections is empty. Edit the user and grant pages.
User cannot send a prompt	No virtual key, or budget exhausted.
Sub-admin sees a page but save returns 403	The sub_admin role catalogue does not include that write operation.
SCIM 403	SCIM disabled or wrong bearer token.
AD test fails	Firewall, bind DN, password, base DN, or TLS mode.
Duplicate person after AD sync	Email matched an existing user with a different username. Merge manually. Sync will not do it.
Guard does not log prompts	Agent offline, PAC not set, or host not in Target Websites.
macOS pkg looks old	Use the zip. Do not ship the stale pkg.

12. Startup, shutdown, and restart

48. Start: docker compose up -d. Expected result: gateway container status is running.
49. Stop: docker compose stop. Expected result: the port no longer accepts connections. The database is unchanged.
50. Restart after a config edit in .env: docker compose up -d --force-recreate. Expected result: the new environment is visible in docker compose exec printenv for a non-secret key such as APP_PORT.
51. Rebuild after code changes: docker compose up -d --build. Take a database backup first on a production host.

Do not delete the ./data volume as a restart step. That directory holds runtime data including config.json. The compose file warns not to bind-mount a single file onto /app/data/config.json because atomic saves then fail.

13. Health checks you can run

52. Container: docker compose ps. Expected result: gateway_tech is Up.
53. Process health: GET http://<host>:<APP_PORT>/health or /api/health. Expected result: JSON with status ok when the process can answer. A database problem is reported inside that payload when DB pings are enabled.
54. Application: sign in as admin in a browser.
55. Model path: one chat call with a virtual key and a known model.
56. Guard path: one online agent on Browser AI, Guard Agents, and one test prompt in Prompt Logs.

A green container is not proof that a provider key works or that browser rules match your sites.

14. SMTP and mail

User creation, password reset, and AD sync send mail through the configured SMTP settings (handlers/smtp.go and tables/smtp.go). If mail is not configured, users are still created but they may not receive the temporary password. Check SMTP in settings before a bulk AD sync. Do not put the SMTP password in this document.

15. Ports

Port	Where it is defined	Exposure
APP_PORT	.env, published by compose	Reachable as configured. Put TLS in front for production.
DB_PORT	.env	Database network only.
PROXY_PORT	.env, only with profile network-proxy	Do not enable unless you intend a shared office proxy.
18103	Laptop Guard MITM	Loopback. Not a server firewall opening.
18195	Laptop PAC	Loopback.

Older Markdown in docs/ mentions additional host ports such as 6000. Those are examples from a specific deployment, not a value compose will choose for you. Use the APP_PORT in your .env.

16. What this manual does not verify

- A live login on your server was not performed while writing this edition.
- A full Go test run was not used as a release gate for this document.
- Mac package signing and notarisation are not verified. The zip version label 1.1.20 is a file fact. A newly signed pkg is a separate Mac build.

- Backup cron, disk encryption, and SIEM forwarding are recommended operations. They are not installed by the repository automatically.

17. Hardening checklist

57. Place the gateway behind TLS. Do not expose database ports on the public internet.
58. Set DB_SSL_MODE to the value your database requires.
59. Keep GATEWAY_GUARD_REQUIRE_SECRET at 1.
60. Use LDAPS for Active Directory.
61. Restrict who can open Roles and Permissions and User Provisioning.
62. Review prompt logs for sensitive data and limit who can export them.
63. Rotate ADMIN_PASSWORD, provider keys, SCIM bearer token, and GATEWAY_GUARD_SECRET on staff changes.

18. Password reset operations

64. Generate a random PASSWORD_RESET_SECRET of at least 32 characters. Store it only in .env or the secret store.
65. Restart the gateway so the process reads it.
66. Use Forgot password on the sign-in page with a mailbox that SMTP can reach.
67. Expected result: the user receives a code or link and can set a new password.
68. If the secret is missing, the server log says OTP reset will return HTTP 500. Fix the secret. Do not debug this by weakening password rules.

19. Browser AI day-to-day operations

19.1 Targets and rules

69. Add a host under Target Websites. Save.
70. Add a Guard Rule. Choose pattern or AI Guard Bot, and an action of block, redact, or warn.
71. Use the product test action for the bot or the regex helper if the screen offers it. Those calls are POST /api/browser-ai/rules/test-bot and generate-regex.
72. From a managed laptop, send a harmless test string that matches the rule.
73. Expected result: Prompt Logs shows the event and the action.

19.2 Agents

74. Open Guard Agents. A healthy laptop heartbeats.
75. Pause stops inspection for that agent. Resume turns it back on. Use pause during a fault, not as a permanent exception.
76. Remote uninstall asks the agent to remove itself. Keep the uninstall key in a password manager. The API can rotate it.
77. Deleting an agent row removes the dashboard record. It is not a substitute for uninstalling the program if the laptop is still running it.

19.3 Log cleanup

78. Prompt Logs and Search Logs support clearing by period: 1 day, 7 days, 30 days, or all.
79. The caller needs Logs Delete. A user with only View will receive 403.
80. Expected result: older rows disappear. Take an export first if audit policy requires a copy.

19.4 Rebuild packages

POST /api/browser-ai/setup/rebuild refreshes setup packages from the server. GET /api/browser-ai/setup/info and rebuild-history show what was produced. Employees should download from Setup, not from a copied USB of an old pkg. Windows zip and macOS zip are the update channels used by the update scripts.

20. Verification commands

```
docker compose ps
curl -sS http://127.0.0.1:<APP_PORT>/health
curl -sS http://127.0.0.1:<APP_PORT>/api/health
```

Expected health body includes status ok when the process is healthy. If database pings are enabled, a database failure is reported in that JSON instead of a silent ok.

```
curl -sS http://127.0.0.1:<APP_PORT>/v1/models -H "Authorization: Bearer <VIRTUAL_KEY>"
```

Expected result: a model list, or a clear provider error. 401 means the virtual key was rejected.

21. Feature flags and section gates

handlers/section_flag_gate.go can gate /api/browser-ai with the flag section.browser-ai. If Browser AI disappears for everyone, including an admin who should see it, check feature flags before you debug the agent.

22. Network proxy profile

81. Set PROXY_PORT and NETWORK_PROXY_CONTAINER_NAME in .env if you use this profile.
82. Start with: docker compose --profile network-proxy up -d
83. The container runs mitmdump with GATEWAY_FAIL_OPEN=0, GATEWAY_SERVER_MODE=1, and agent type network.
84. Point a lab PAC at https://<backend>/api/browser-ai/pac only if you intentionally want a shared proxy.
85. Do not enable this profile on a production laptop fleet. The compose comment says the employee path is the Guard program on each laptop.

23. Directory operations checklist

Step	Cloud SCIM	On-premises AD
Identity match	external id, email, or user name	username only

Default role	user	user, unless default_role is changed in stored config
Teams	IdP groups become teams	SCIM default team only, if set
Virtual key	not assigned	not assigned
Pages	not assigned	not assigned
Duplicate email	existing user is reused	a second user can be created

24. Operational risks

- DB_SSL_MODE default disable sends database traffic without TLS until you change it.
- The gateway container is started as user 0:0 in compose. Treat the host Docker socket and the .env file as high privilege.
- Skill marketplace URLs are public. Review skill text before you publish.
- Prompt logs store employee text. Limit who has Logs Read and Browser AI pages.
- An AI Guard Bot sends extracted text to the configured model. That is another disclosure. Prefer regex for known secret formats when a model call is not acceptable.
- False negatives: a site that is not a target, a parser that misses the text box, or a rule that does not match.
- False positives: a regex that is too wide, or a model that flags ordinary language.

End of system manual.